
Kahlus HNPH: A Falsifiable Protocol for an Oracle-Conditioned Sleep-Transition Predictability Frontier

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Abstract

Most automated sleep studies ask which stage a person is in now. HNPH asks a harder but narrower question: given an oracle annotation of the current stage, does EEG recorded before an issue time improve the probability assigned to a stable future annotation beyond what persistence, recent stage history, and time of night already predict? A high forecasting score alone cannot answer this question because it may come from overlapping windows, subject identity, acquisition artifacts, or a weak comparator. We present version 0.4 of the Human Neural Predictability Horizon (HNPH), a falsifiable protocol for estimating the longest contiguous set of future lead bands with residual EEG information. The outcome is an *operational sleep-stage annotation conditioned on oracle current state*, not a biological transition detector. The planned study uses DOD-H for development and keeps DOD-O sealed for external acquisition/population evaluation, conditional on a fail-closed audit of five individual rater streams, person identities, physical metadata, licenses, and immutable hashes. The protocol prespecifies leave-one-rater-out soft targets, person-first splits, disjoint physical-time support, subject-balanced proper-score inference, best-baseline gates, negative controls, and powered simultaneous confidence bounds. Sleep-EDF and CAP are limited to descriptive single-label transport checks. This is a protocol/theory paper: source qualification is unverified, every empirical stage is unrun, and no HNPH frontier is claimed.

Scientific status (13 July 2026). Protocol v0.4 is frozen, but DOD source qualification is **UNVERIFIED**. No training, model selection, DOD-O opening, or empirical HNPH claim is authorized. Every schematic is visibly labeled. The sole signal panel is a descriptive, single-label Sleep-EDF transport example and is not HNPH evidence.

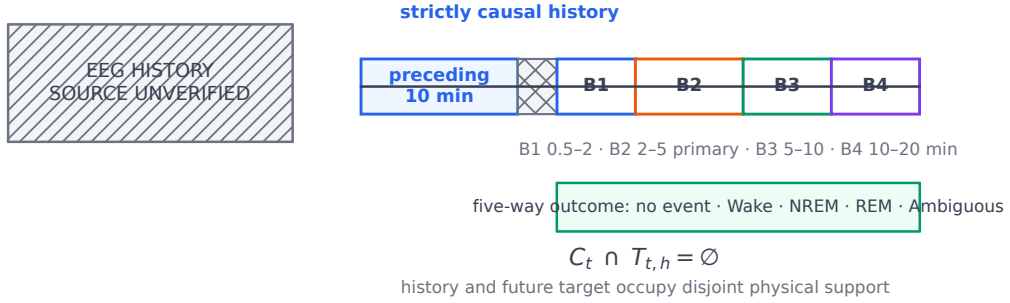
1 Scope and motivation

Sleep classification and sleep forecasting are different tasks. A classifier estimates the present stage from a present window. A forecaster must use only information available before an issue time to estimate a later outcome. HNPH makes that boundary explicit: after current scored state, recent state history, bout age, elapsed night time, and declared acquisition nuisance are accounted for, does strictly causal EEG history improve a probability distribution for a future stable sleep-stage annotation? Figure 1 shows the operational task.

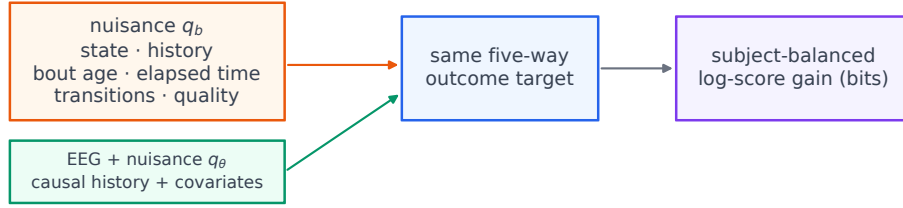
The distinction matters because sleep is persistent. A model can look impressive by predicting that the current stage will continue, or by seeing nearly the same samples in both its input and target. HNPH therefore supplies the issue-time state as an oracle annotation and asks only whether EEG adds residual information beyond a strong nuisance model. Even a positive result would concern the

Sleep-transition forecasting is anchored on causal history, not future labels

A. CAUSAL OBSERVATION AND PHYSICAL TIME



B. ISOLATING INCREMENTAL EEG INFORMATION



HNPH v0.4 · protocol_sha256PROTOCOL SCHEMATIC SOURCE UNVERIFIED NO PERFORMANCE RESULT no empirical frontier claim

Figure 1: Operational HNPH task. An oracle current-stage annotation and causal EEG history precede a guard interval and four future lead bands. The five-way leave-one-rater-out target is scored against the best eligible nuisance comparator. This is a task schematic, not a performance result.

specified annotation protocol; it would not detect an unobserved biological transition or establish deployment performance.

This restriction is also an explicit correction to the previous Kahlus draft. Its 127-sample target overlapped the input by 126 samples, and the reported value came from a GRU translator rather than the proposed Neural Field Compiler. The often-quoted 3.116 MSE is therefore retained only as an invalidated historical sidecar result. It is not HNPH evidence, and no plot or metric from that experiment is reused here.

Sleep-stage prediction and forecasting are not new. U-Sleep demonstrated cross-cohort present-stage classification at scale [8]; SSF-SET forecasts future sleep-stage sequences from past EEG [7]. HNPH’s proposed contribution is instead the nuisance-standardized estimand and evidence contract: a strong state-duration comparator, person-balanced inference, repeated-rater construct audit, an untouched external cohort, and controls evaluated under one gate. A positive, null, blocked, or invalid study must remain distinguishable.

Architectural names, geometry-aware operators, future hardware, disease applications, and passive-PCI hypotheses are outside this protocol. They require later, separately frozen protocols. Version 0.4 evaluates a causal probabilistic EEG forecaster, not NFC and not a clinical system.

2 Operational target and estimand

For person s and issue time t , let H_t be EEG supported strictly on $[t - L, t)$ and let Z_t contain only declared causal nuisance variables: oracle current macrostate, two preceding states, current bout age, elapsed recording time, recent transition count, and signal-quality features. Subject, site, path, session, and future-label identifiers are forbidden model inputs. Wake, NREM, REM, and unknown

annotations are evaluated on the native 30-second grid. A destination transition is stable after two consecutive destination epochs.

For lead band B , the five-way outcome is

$$O_t^B \in \{\text{no event, Wake, NREM, REM, Ambiguous}\}.$$

The preregistered bands are $(0.5, 2]$, $(2, 5]$, $(5, 10]$, and $(10, 20]$ minutes; $(2, 5]$ is primary. Unknown intervals and rater disagreement are scored as Ambiguous rather than silently discarded. Incomplete follow-up is excluded and counted by band.

Let $q_b(o \mid Z_t)$ be the best eligible nuisance distribution selected on development data and $q_\theta(o \mid H_t, Z_t)$ an EEG-augmented distribution. The subject-level gain is

$$G_s(B) = \frac{1}{|\mathcal{A}_s|} \sum_{t \in \mathcal{A}_s} \log_2 \frac{q_\theta(O_t^B \mid H_t, Z_t)}{q_b(O_t^B \mid Z_t)}, \quad (1)$$

where $\mathcal{A}_s$ contains complete natural-grid anchors. The population estimand $\Delta(B) = \mathbb{E}_s[G_s(B)]$ therefore gives equal weight to people, not windows. Log score is strictly proper and evaluates the full predictive distribution [3]. The empirical frontier is the longest contiguous lead-band prefix for which the simultaneous one-sided 95% subject-cluster lower bound exceeds the preregistered 0.02-bit design threshold and every calibration/control gate passes. An isolated later band cannot skip a failed earlier band.

3 Fail-closed source and label qualification

The Dream Open Datasets publication describes DOD-H and DOD-O as openly distributed polysomnography cohorts independently scored by five sleep technologists [4]. The project repository and versioned archive are treated only as candidate official sources until locally hash-qualified [2, 5]. Protocol v0.4 assigns DOD-H to development and DOD-O to sealed external acquisition/population evaluation. That assignment is conditional, not an assertion about the contents of any local download.

Before migration, a local qualification packet must verify dataset/version, license, immutable archive hash, stable person identity, channel-level physical units and sampling rates, and five separately identifiable annotation streams with per-rater hashes. If individual labels are absent, the migration stops and HNPH remains protocol-only. Consensus labels cannot replace the repeated-rater gate. Sleep-EDF and CAP may support explicitly descriptive transport plots, but their single-label role under this protocol cannot enable construct-validity claims [6, 9].

For each target rater r , the soft target is computed only from other raters; r must not appear among its consensus contributors. At least three independent contributing raters are required at every scored target. The target artifact binds target-rater identity, contributing-rater identities, annotation hashes, stage mapping, and target hash. Figure 2 summarizes the full leakage boundary.

4 Study design and decision gates

People are split before anchors, transforms, calibration, baseline selection, or model fitting. Every input and target interval is represented in physical time and must be disjoint, including filter support and guard intervals. DOD-O remains inaccessible until the protocol, model family, seeds, calibration method, comparators, report template, and source qualification are frozen.

The baseline ladder is empirical no-event/destination frequency, Markov prior, semi-Markov competing risk, validation-tuned logistic or ridge hazard, validation-tuned gradient-boosted hazard, and fixed standard EEG features plus nuisance. Any candidate must beat the best eligible comparator, not merely a convenient weak baseline. The semi-Markov comparator must pass dwell-time fit, held-person calibration and Brier decomposition, synthetic no-EEG dominance, simpler-Markov comparisons, and a flexible nuisance-challenger audit.

Required controls are label shuffle, within-person time shift, unrelated context, missingness-only prediction, applicable channel/geometry permutation, cluster-permutation null, a nuisance probe, a known-signal positive control, and a synthetic null that must fail. Confidence intervals use at least 2,000 person-cluster bootstrap replicates with max- t multiplicity control. Feasibility requires at

A. PHYSICAL-TIME FORECAST ANCHOR

complete follow-up required · native 30-second grid

all recordings from one person remain in one partition

HNPH v0.4 · protocol_sha256=401a8e471d03a15c49401400 · source_qualification=empirical evidence is schematic — no empirical frontier claim

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[illegible]

Table 1: Protocol-release status. Passing software tests does not establish an empirical HNPH result.

Component	Status	Claim consequence
HNPH v0.4 protocol and hash-bound addendum	FROZEN	Supersedes v0.3 for future claim-mode work.
Figure renderer, manifest, and refusal tests	FROZEN	Schematics reproducible; no result implied.
DOD-H/DOD-O local source qualification	BLOCKED	Dataset migration and training unauthorized.
Five-way leave-one-rater-out target packet	UNRUN	Construct-validity gate unavailable.
Baseline, power, comparator, and control gates	UNRUN	No model family authorized.
Sealed DOD-O evaluation	UNRUN	No external or frontier claim.

The next action is qualification, not training. The official archives must be acquired locally; the five annotation streams, identities, units, rates, licenses, and hashes must be checked; and the audit must emit a subject-safe qualification packet. If qualification passes, DOD-H people and leave-one-rater-out targets can be built and the baseline, power, and control gates can run. Only a passing feasibility packet authorizes a frozen small model family. DOD-O remains sealed throughout.

HNPH v0.4 is a falsifiable protocol for an oracle-conditioned operational annotation frontier. It neither claims a measured horizon nor mistakes current-state persistence for future information. Its scientific value will come from making a supported frontier, a bounded null, a blocked construct, and an invalid execution separately auditable. At present the honest outcome is blocked source qualification. Figure polish does not change that status.

[illegible]

A Raw illustration and provenance

Figure 6 shows the raw-to-evidence path. Local raw material is admitted only through source qualification. Person-first manifests, causal target builders, frozen comparisons, and the gate each emit hash-bound artifacts. The figure manifest records the protocol, inputs, classification, panels, outputs, command, and claim scope.

Let $p_1(o) = p(o \mid H, Z)$ and $p_0(o) = p(o \mid Z)$. For oracle distributions,

This identity follows directly from conditional mutual information. For fitted distributions,

$$+ \frac{1}{\ln 2} \mathbb{E} \text{KL}(p_0 \| q_b). \quad (4)$$

Sleep-EDF Expanded · record SC4002E0 · descriptive single-label transport example

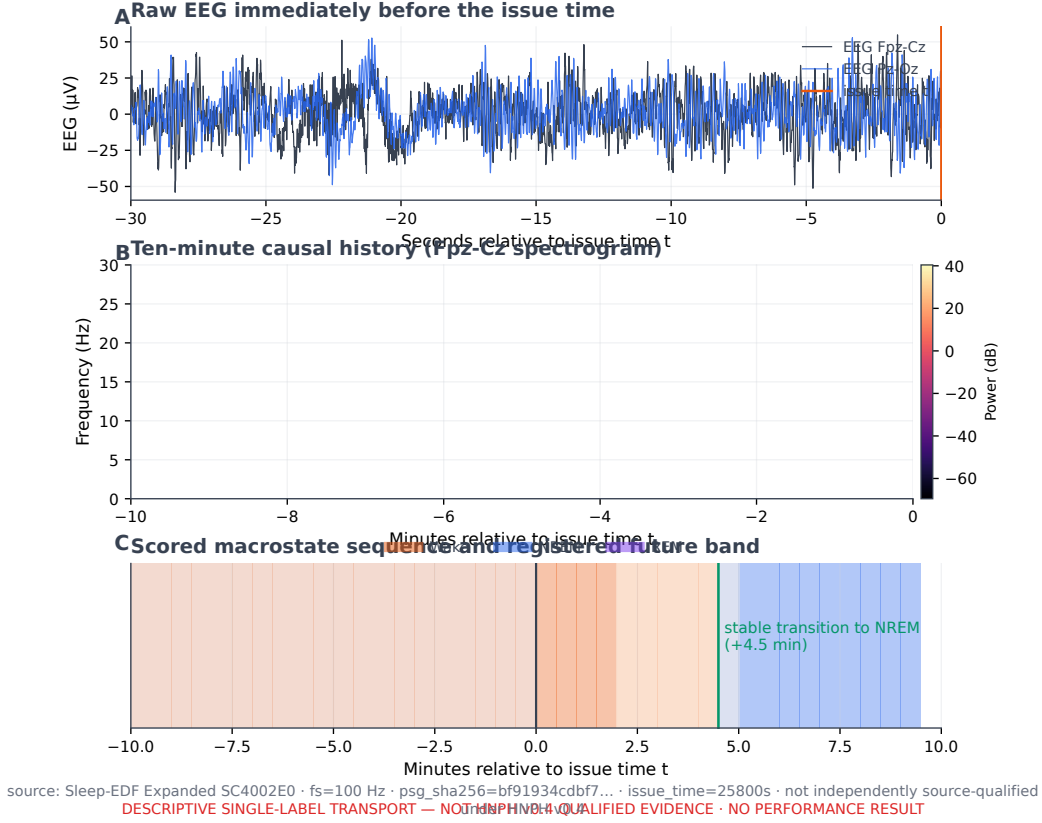


Figure 5: Descriptive Sleep-EDF trace, spectrogram, and single-label hypnogram from record SC4002E0. The origin package reports 100 Hz sampling and hash-bound PSG/hypnogram sources. This transport illustration is not independently source-qualified under HNPH v0.4 and cannot enable a repeated-rater construct-validity or empirical frontier claim.

Positive fitted gain can therefore reflect baseline misspecification, which is why the comparator ladder is part of the scientific gate.

A nontrivial universal finite horizon does not follow from noisy noninvasive observation alone. If an explicitly specified latent process U_t generates observed EEG and future annotations through an appropriate conditional channel, data processing may yield

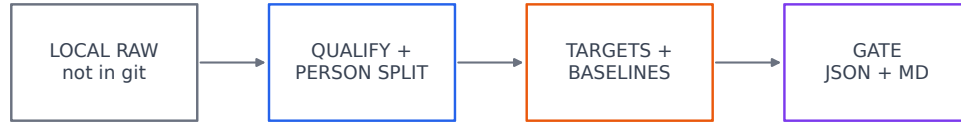
$$I(O_{t+h}; H_t | Z_t) \leq I(U_{t+h}; U_{\leq t} | Z_t).$$

Additional mixing or strong data-processing assumptions can make the right-hand side contract with h [1]. No contraction constants are estimated here. Figure 7 therefore labels the ceiling as a standard, assumption-dependent consequence and not an empirical or novel contribution.

References

- [1] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley, 2 edition, 2006.
- [2] Dreem Organization. Dreem learning open: Official dreem open datasets repository, 2026. URL <https://github.com/Dreem-Organization/dreem-learning-open>. Accessed 13 July 2026.
- [3] Tilman Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.

Hash-bound lineage keeps raw neural data out of git while preserving auditability



SHA-256 binds every boundary; emitted artifacts contain subject-safe IDs only

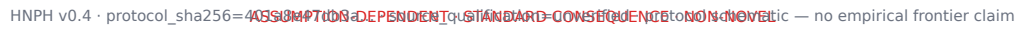
runner → gate → JSON + Markdown

HNPH v0.4 · protocol_sha256=401a8e... **PROVENANCE SCHEMATIC — NO EMPIRICAL EVIDENCE** schematic — no empirical frontier claim

Figure 6: Raw-to-evidence lineage. Hash-bound local sources feed qualification, person-first manifests, causal targets, frozen comparisons, and paired JSON/Markdown evidence; raw neural samples remain outside git.

- [4] Antoine Guillot, Fabien Sauvet, Emmanuel H. During, and Valentin Thorey. Dreem open datasets: Multi-scored sleep datasets to compare human and automated sleep staging. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 28(9):1955–1965, 2020. doi: 10.1109/TNSRE.2020.3011181.
- [5] Antoine Guillot, Fabien Sauvet, Emmanuel H. During, and Valentin Thorey. Dreem open datasets, 2025.
- [6] Bob Kemp. Sleep-EDF Database Expanded, version 1.0.0, 2018. URL <https://physionet.org/content/sleep-edfx/1.0.0/>.
- [7] Young-Seok Kweon, Gi-Hwan Shin, Dae-Hyeok Lee, and Seong-Whan Lee. SSF-SET: A discrete eeg token-based framework for sleep stage forecasting. *IEEE Journal of Biomedical and Health Informatics*, 2026. doi: 10.1109/JBHI.2026.3662787. Online ahead of print.
- [8] Mathias Perslev, Sune Darkner, Lykke Kempfner, Miki Nikolic, Poul Jorgen Jennum, and Christian Igel. U-Sleep: Resilient high-frequency sleep staging. *npj Digital Medicine*, 4:72, 2021. doi: 10.1038/s41746-021-00440-5.
- [9] PhysioNet. CAP Sleep Database, version 1.0.0, 2012. URL <https://physionet.org/content/capslpdb/1.0.0/>.

Observation contraction



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